

KON 313

Feedback Control Systems

Modeling of the Electrical Networks in Frequency Domain

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- Passive Elements in Electrical Circuits
- Operational Amplifiers

Passive Elements in electrical Circuits

- Passive elements can store energy and release the stored energy into the system. They do not have an external power supply. However, they can only deliver the energy stored previously.
- The electric networks that we work with consist of three passive linear components:

1) resistors, $v(t) = Ri(t)$

2) capacitors, $v(t) = \frac{1}{C} \int_0^t i(\tau) d\tau$

3) Inductors, $v(t) = L \frac{di(t)}{dt}$

Passive Elements in electrical Circuits

The basic method. The basic alternative method to find the transform function of an electrical network is:

Step 1: Decide on the input and the output

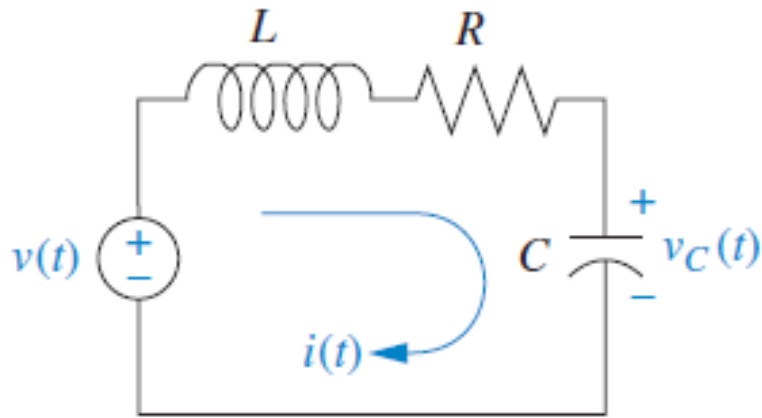
Step 2: Write the differential equations for the circuit

Step 3: Take the Laplace transforms of the differential equations

Step 4: Solve for the transfer function (assuming zero initial conditions).

Passive Elements in electrical Circuits

E.g. Series RLC Circuit, via differential equations: Find the transfer function relating the capacitor voltage, $V_C(s)$, to the input voltage, $V(s)$.



$$v(t) = L \frac{di(t)}{dt} + Ri(t) + \frac{1}{C} \int_0^t i(\tau) d\tau$$

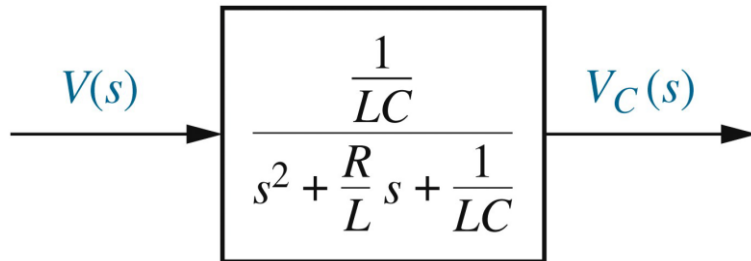
$$v_C(t) = \frac{1}{C} \int_0^t i(\tau) d\tau \Rightarrow i(t) = C \frac{dv_C(t)}{dt}$$

$$\Rightarrow v(t) = LC \frac{d^2 v_C(t)}{dt^2} + RC \frac{dv_C(t)}{dt} + v_C(t)$$

$$\Rightarrow V(s) = LCs^2 V_C(s) + RCs V_C(s) + V_C(s)$$

$$\Rightarrow V(s) = (LCs^2 + RCs + 1) V_C(s)$$

$$\Rightarrow \frac{V_C(s)}{V(s)} = \frac{\frac{1}{LC}}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$



Passive Elements in electrical Circuits

- Passive elements in s -domain:

1) resistors, $v(t) = Ri(t) \implies V(s) = RI(s)$

2) capacitors, $v(t) = \frac{1}{C} \int_0^t i(\tau) d\tau \implies V(s) = \frac{1}{Cs} I(s)$

3) Inductors, $v(t) = L \frac{di(t)}{dt} \implies V(s) = LsI(s)$

- **Impedance.** This function is similar to the definition of resistance, that is, the ratio of voltage to current. But, unlike resistance, this function is applicable to capacitors and inductors and carries information on the dynamic behavior of the component, since it represents an equivalent differential equation.

$$\frac{V(s)}{I(s)} = Z(s)$$

Passive Elements in electrical Circuits

- Passive elements

TABLE 2.3 Voltage-current, voltage-charge, and impedance relationships for capacitors, resistors, and inductors

Component	Voltage-current	Current-voltage	Voltage-charge	Impedance $Z(s) = V(s)/I(s)$	Admittance $Y(s) = I(s)/V(s)$
 Capacitor	$v(t) = \frac{1}{C} \int_0^t i(\tau) d\tau$	$i(t) = C \frac{dv(t)}{dt}$	$v(t) = \frac{1}{C} q(t)$	$\frac{1}{Cs}$	Cs
 Resistor	$v(t) = Ri(t)$	$i(t) = \frac{1}{R} v(t)$	$v(t) = R \frac{dq(t)}{dt}$	R	$\frac{1}{R} = G$
 Inductor	$v(t) = L \frac{di(t)}{dt}$	$i(t) = \frac{1}{L} \int_0^t v(\tau) d\tau$	$v(t) = L \frac{d^2q(t)}{dt^2}$	Ls	$\frac{1}{Ls}$

Note: The following set of symbols and units is used throughout this book: $v(t)$ – V (volts), $i(t)$ – A (amps), $q(t)$ – Q (coulombs), C – F (farads), R – Ω (ohms), G – Ω (mhos), L – H (henries).

Passive Elements in electrical Circuits

The simplifying technique. The technique for simplifying the solution to find the transform function of electrical networks is summarized as:

Step 1: Decide on the input and the output

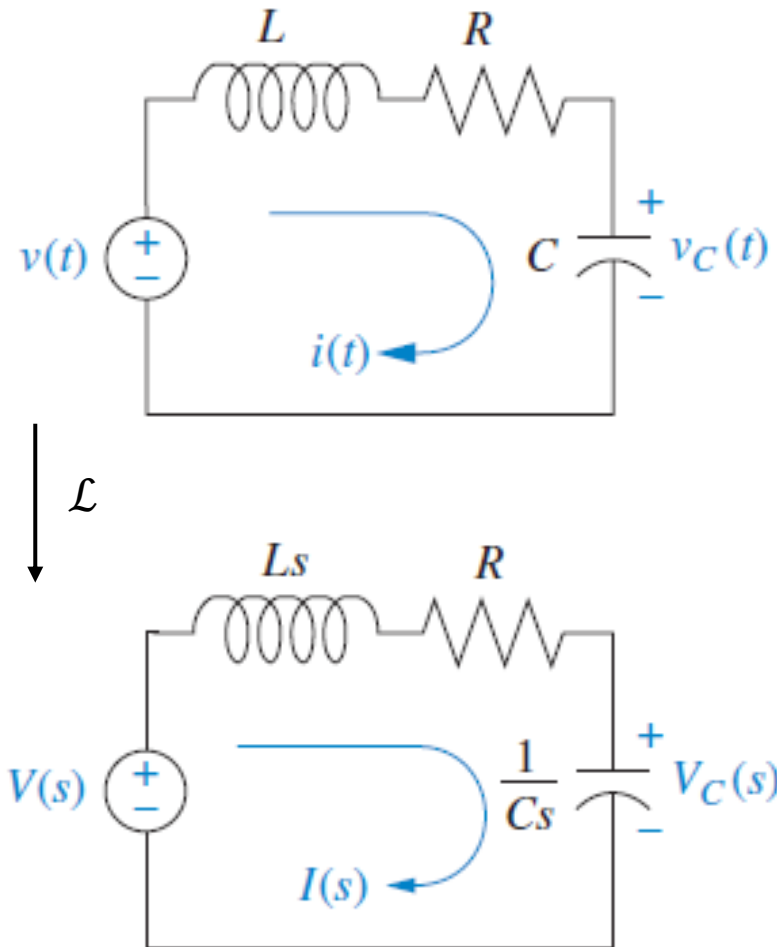
Step 2: Redraw the circuit in Laplace domain (represents all of the elements in Laplace domain, assuming zero initial conditions)

Step 3: Use the required electrical impedance laws to achieve the equations (Ohm, KVL, KCL, Node-voltage, mesh-current, parallel/series, Δ/Y , Y/Δ ,

Step 4: Solve for the transfer function.

Passive Elements in electrical Circuits

E.g. Series RLC Circuit, via Laplace transform, without differential equations: Find the transfer function relating the capacitor voltage, $V_C(s)$, to the input voltage, $V(s)$.



Voltage divider:

$$\begin{aligned}\Rightarrow \frac{V_C(s)}{V(s)} &= \frac{\frac{1}{Cs}}{Ls + R + \frac{1}{Cs}} \\ &= \frac{\frac{1}{LC}}{s^2 + \frac{R}{L}s + \frac{1}{LC}}\end{aligned}$$

Passive Elements in electrical Circuits

E.g. Multiple Meshes, via Laplace transform:

For the given network, find the transfer function relating $I_2(s)$ to $V(s)$.

Mesh-current:

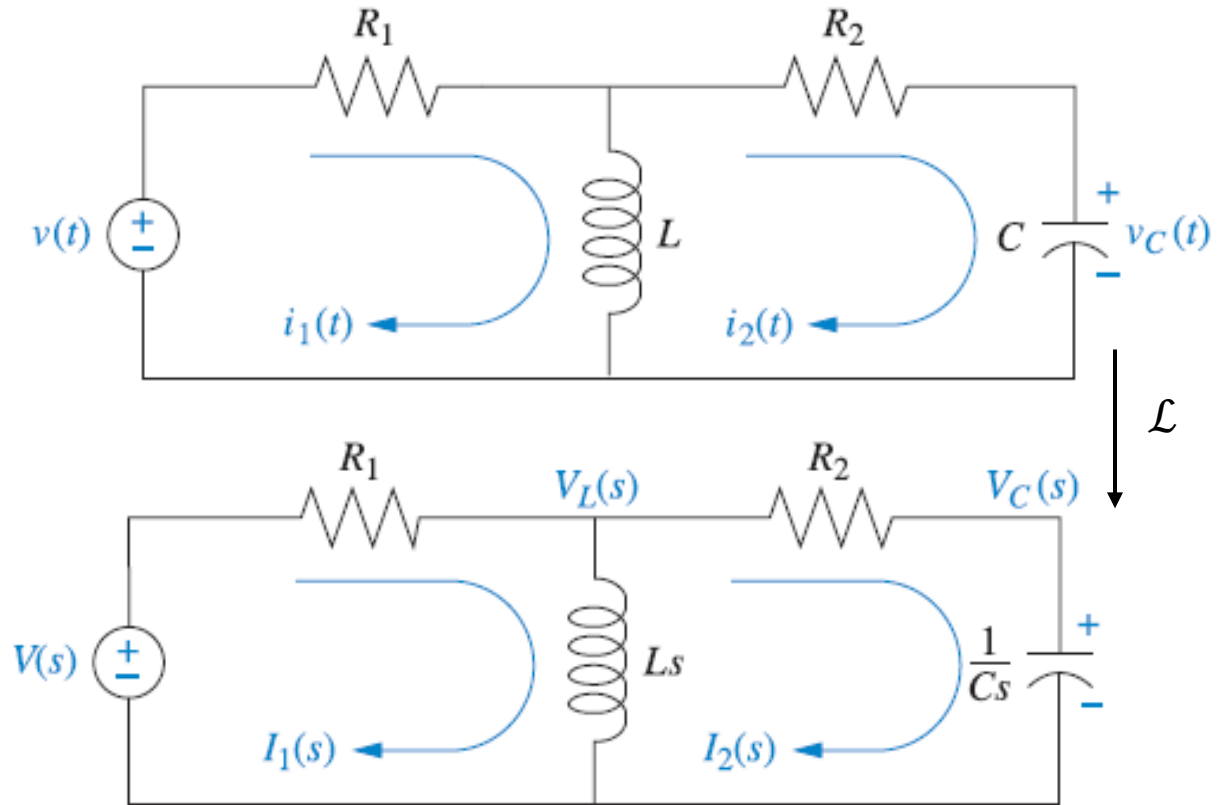
KVL1: $R_1 I_1(s) + Ls I_1(s) - Ls I_2(s) = V(s)$

KVL2: $Ls I_2(s) + R_2 I_2(s) + \frac{1}{Cs} I_2(s) - Ls I_1(s) = 0$

$$(R_1 + Ls)I_1(s) - LsI_2(s) = V(s)$$

$$-LsI_1(s) + \left(Ls + R_2 + \frac{1}{Cs} \right) I_2(s) = 0$$

You can use Cramer's rule (or any other method for solving simultaneous equations) to solve these equations.



$$\frac{V(s)}{I_2(s)} = \frac{LCs^2}{(R_1 + R_2)LCs^2 + (R_1R_2C + L)s + R_1}$$

Passive Elements in electrical Circuits

E.g. Multiple Meshes, via Laplace transform:

$$(R_1 + Ls)I_1(s) - LsI_2(s) = V(s)$$

$$-LsI_1(s) + \left(Ls + R_2 + \frac{1}{Cs}\right)I_2(s) = 0$$

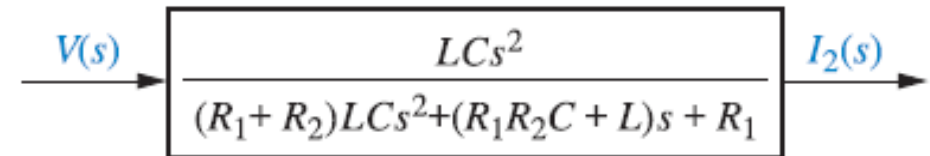
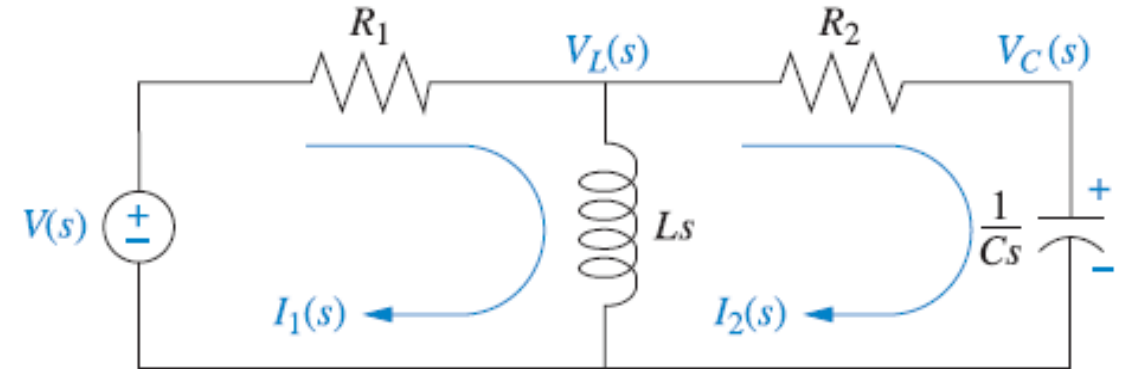
Cramer's rule

$$I_2(s) = \frac{\begin{vmatrix} (R_1 + Ls) & V(s) \\ -Ls & 0 \end{vmatrix}}{\Delta} = \frac{LsV(s)}{\Delta}$$

$$\Delta = \begin{vmatrix} (R_1 + Ls) & -Ls \\ -Ls & \left(Ls + R_2 + \frac{1}{Cs}\right) \end{vmatrix}$$



$$G(s) = \frac{I_2(s)}{V(s)} = \frac{Ls}{\Delta} = \frac{LCs^2}{(R_1 + R_2)LCs^2 + (R_1R_2C + L)s + R_1}$$



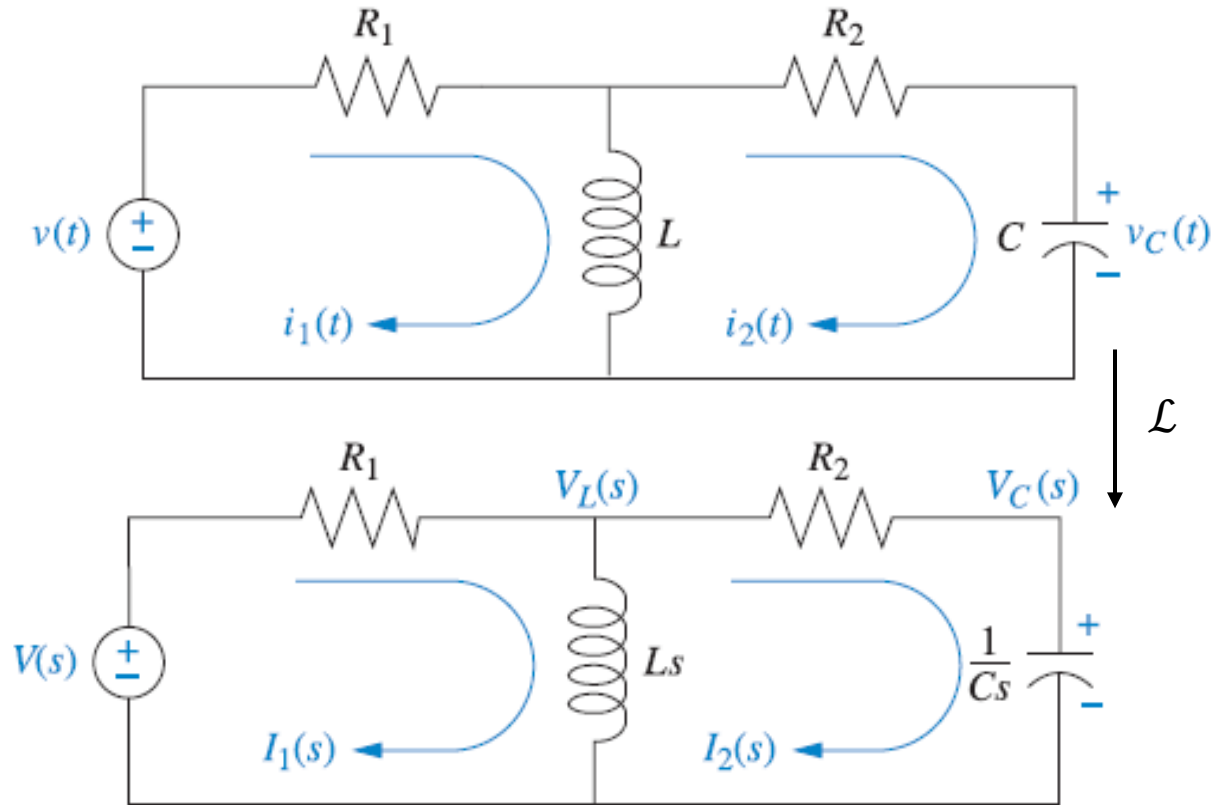
Passive Elements in electrical Circuits

E.g. Multiple Meshes, via Laplace transform: Alternative technique for mesh-current method:

$$\begin{aligned} \left[\begin{array}{c} \text{Sum of} \\ \text{impedances} \\ \text{around Mesh 1} \end{array} \right] I_1(s) - \left[\begin{array}{c} \text{Sum of} \\ \text{impedances} \\ \text{common to the} \\ \text{two meshes} \end{array} \right] I_2(s) &= \left[\begin{array}{c} \text{Sum of applied} \\ \text{voltages around} \\ \text{Mesh 1} \end{array} \right] \\ - \left[\begin{array}{c} \text{Sum of} \\ \text{impedances} \\ \text{common to the} \\ \text{two meshes} \end{array} \right] I_1(s) + \left[\begin{array}{c} \text{Sum of} \\ \text{impedances} \\ \text{around Mesh 2} \end{array} \right] I_2(s) &= \left[\begin{array}{c} \text{Sum of applied} \\ \text{voltages around} \\ \text{Mesh 2} \end{array} \right] \end{aligned}$$

$$R_1 I_1(s) + Ls I_1(s) - Ls I_2(s) = V(s)$$

$$Ls I_2(s) + R_2 I_2(s) + \frac{1}{Cs} I_2(s) - Ls I_1(s) = 0$$

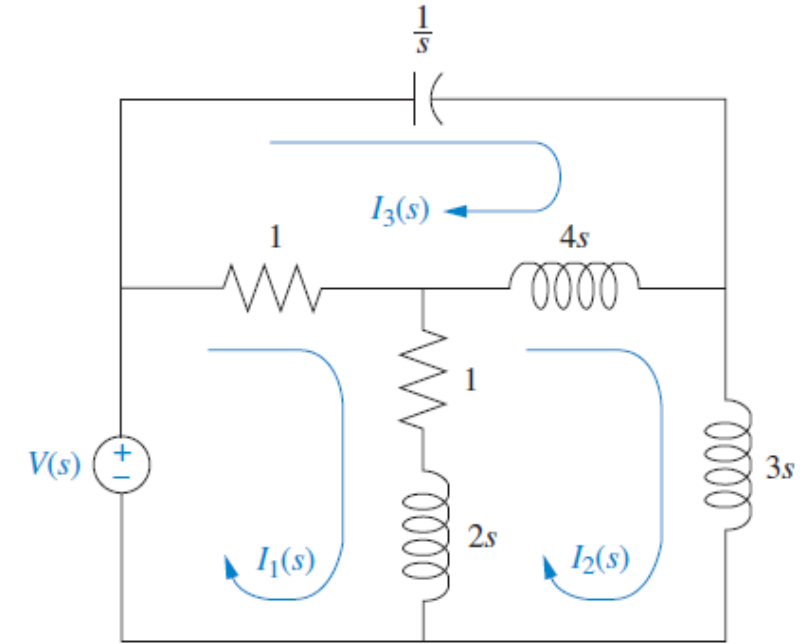


$$\begin{array}{c} \xrightarrow{V(s)} \boxed{\frac{LCs^2}{(R_1 + R_2)LCs^2 + (R_1R_2C + L)s + R_1}} \xrightarrow{I_2(s)} \end{array}$$

Passive Elements in electrical Circuits

E.g. Multiple Meshes, via Laplace transform: Find the transfer function relating

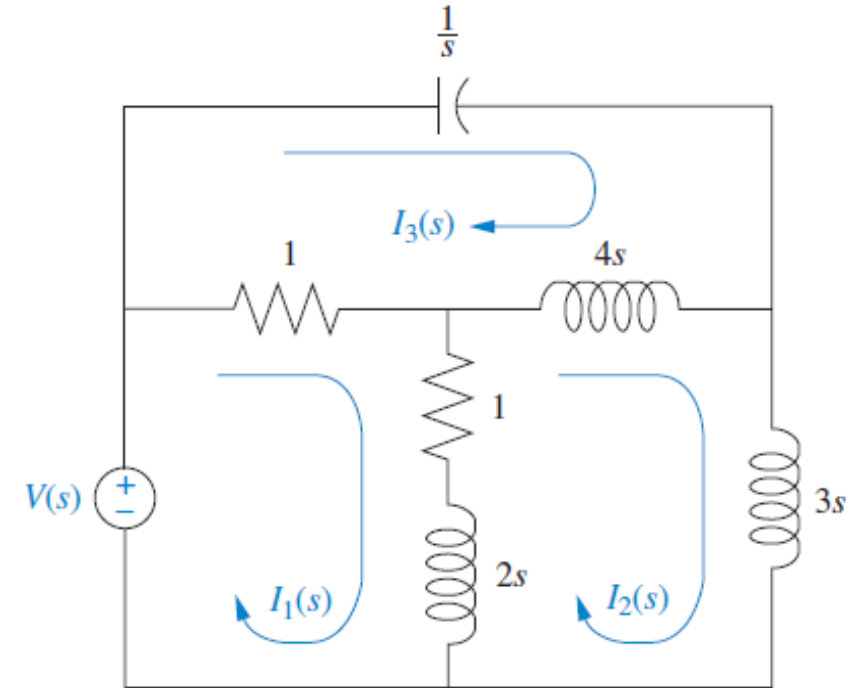
$$\begin{aligned}
 &\left[\begin{array}{c} \text{Sum of} \\ \text{impedances} \\ \text{around Mesh 1} \end{array} \right] I_1(s) - \left[\begin{array}{c} \text{Sum of} \\ \text{impedances} \\ \text{common to} \\ \text{Mesh 1 and} \\ \text{Mesh 2} \end{array} \right] I_2(s) - \left[\begin{array}{c} \text{Sum of} \\ \text{impedances} \\ \text{common to} \\ \text{Mesh 1 and} \\ \text{Mesh 3} \end{array} \right] I_3(s) = \left[\begin{array}{c} \text{Sum of applied} \\ \text{voltages around} \\ \text{Mesh 1} \end{array} \right] \\
 &- \left[\begin{array}{c} \text{Sum of} \\ \text{impedances} \\ \text{common to} \\ \text{Mesh 1 and} \\ \text{Mesh 2} \end{array} \right] I_1(s) + \left[\begin{array}{c} \text{Sum of} \\ \text{impedances} \\ \text{around Mesh 2} \end{array} \right] I_2(s) - \left[\begin{array}{c} \text{Sum of} \\ \text{impedances} \\ \text{common to} \\ \text{Mesh 2 and} \\ \text{Mesh 3} \end{array} \right] I_3(s) = \left[\begin{array}{c} \text{Sum of applied} \\ \text{voltages around} \\ \text{Mesh 2} \end{array} \right] \\
 &- \left[\begin{array}{c} \text{Sum of} \\ \text{impedances} \\ \text{common to} \\ \text{Mesh 1 and} \\ \text{Mesh 3} \end{array} \right] I_1(s) - \left[\begin{array}{c} \text{Sum of} \\ \text{impedances} \\ \text{common to} \\ \text{Mesh 2 and} \\ \text{Mesh 3} \end{array} \right] I_2(s) + \left[\begin{array}{c} \text{Sum of} \\ \text{impedances} \\ \text{around Mesh 3} \end{array} \right] I_3(s) = \left[\begin{array}{c} \text{Sum of applied} \\ \text{voltages around} \\ \text{Mesh 3} \end{array} \right]
 \end{aligned}$$



Passive Elements in electrical Circuits

E.g. Multiple Meshes, via Laplace transform: Find the transfer function relating

$$\begin{aligned} + (2s + 2)I_1(s) - (2s + 1)I_2(s) - I_3(s) &= V(s) \\ -(2s + 1)I_1(s) + (9s + 1)I_2(s) - 4sI_3(s) &= 0 \\ -I_1(s) - 4sI_2(s) + \left(4s + 1 + \frac{1}{s}\right)I_3(s) &= 0 \end{aligned}$$



Operational Amplifiers (OP Amps)

Op amps are frequently used in controllers and for compensation purposes.

Ideal Op Amp has the following characteristics:

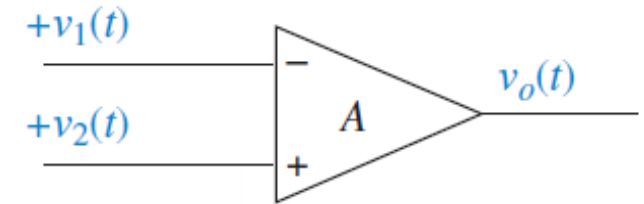
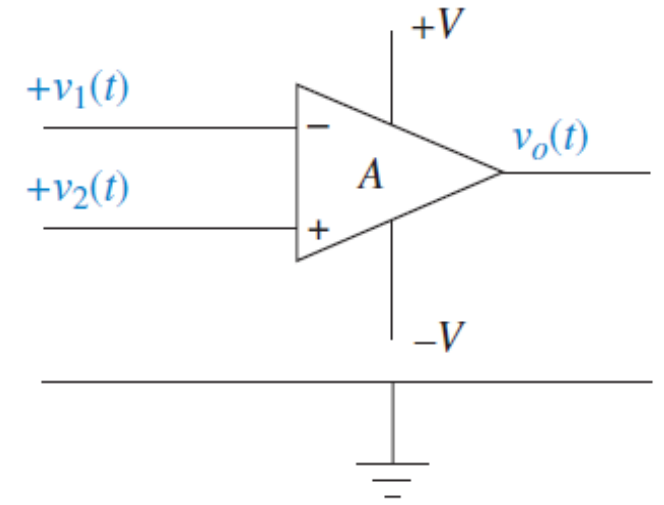
1. High input impedance, $Z_i \rightarrow \infty$
2. Low output impedance, $Z_o \rightarrow 0$
3. High constant gain amplification, $A \rightarrow \infty$

$$\Rightarrow v_1 = v_2$$

$$i_1 = i_2 = 0$$

The output, v_o , is given by:

$$v_o(t) = A(v_2(t) - v_1(t))$$



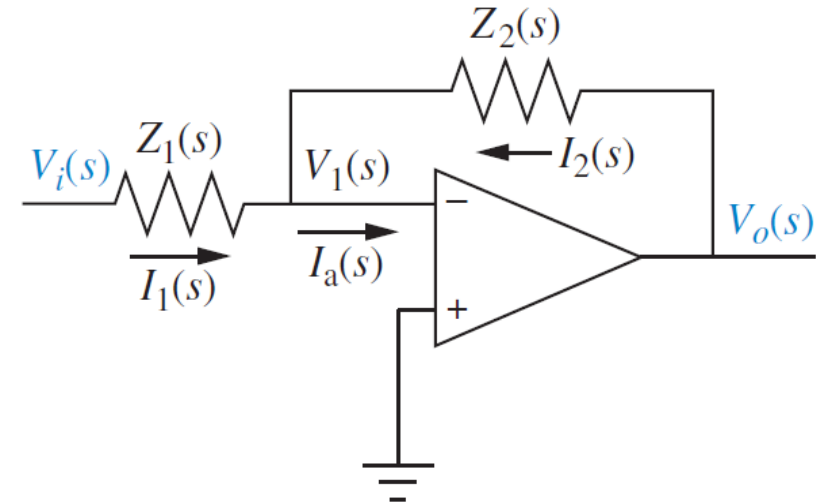
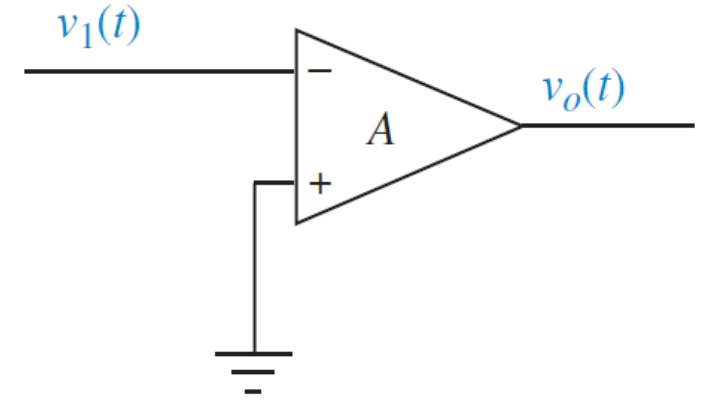
Inverting Operational Amplifiers

If $v_2(t)$ is grounded, the amplifier is called an inverting operational amplifier.

$$v_o(t) = -Av_1(t)$$

If two impedances are connected to the inverting operational amplifier, the transfer function is:

$$\frac{V_o(s)}{V_i(s)} = -\frac{Z_2(s)}{Z_1(s)}$$



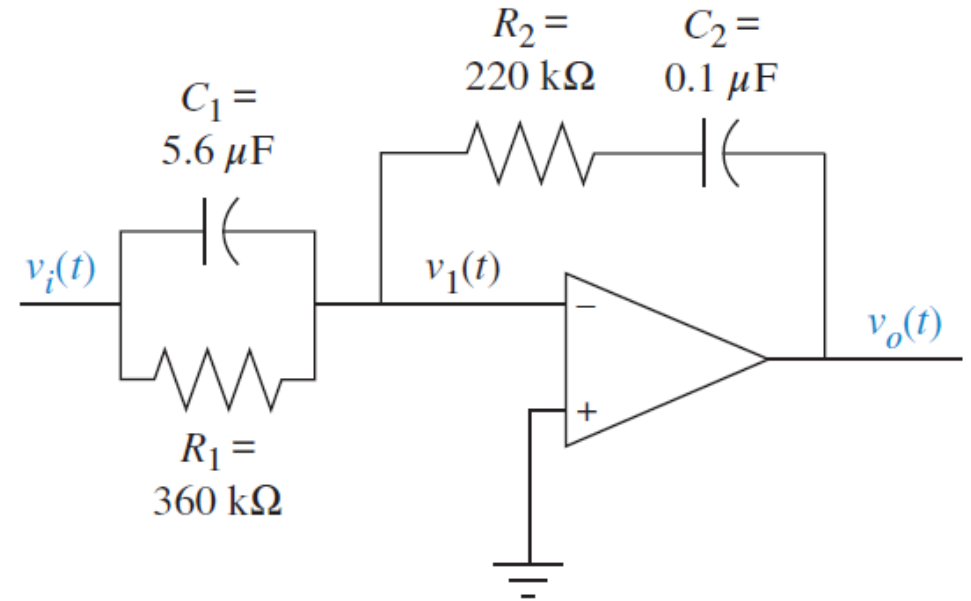
Inverting Operational Amplifiers

E.g. Find the transfer function.

$$Z_1(s) = \frac{1}{C_1 s + \frac{1}{R_1}} = \frac{1}{5.6 + 10^{-6}s + \frac{1}{360 \times 10^3}} = \frac{360 \times 10^3}{2.016s + 1}$$

$$Z_2(s) = R_2 + \frac{1}{C_2 s} = 220 \times 10^3 + \frac{10^7}{s}$$

$$\frac{V_o(s)}{V_i(s)} = -1.232 \frac{s^2 + 45.95s + 22.55}{s}$$



The resulting circuit is called a PID controller and can be used to improve the performance of a control system.

Noninverting Operational Amplifiers

Another circuit that can be analyzed for its transfer function is the noninverting operational amplifier circuit.

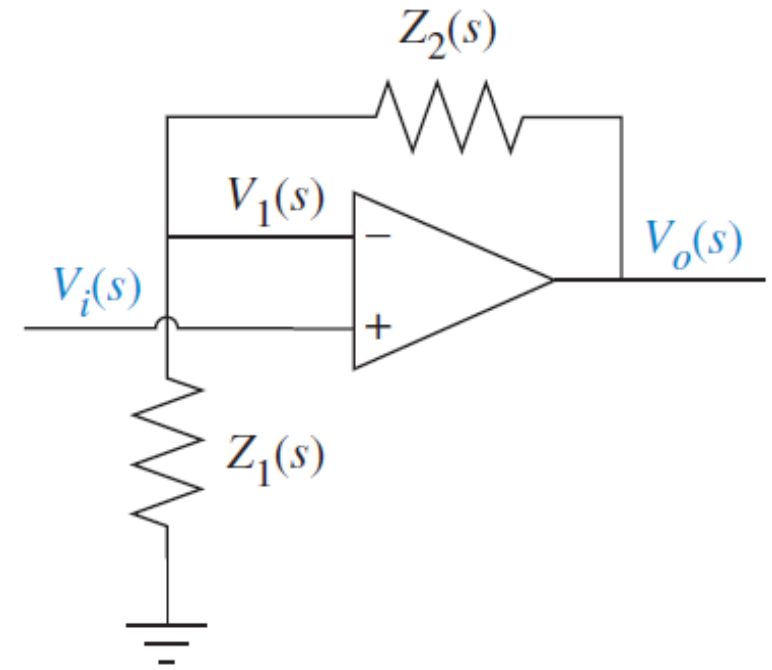
$$V_o(s) = A(V_i(s) - V_1(s))$$

$$V_1(s) = \frac{Z_1(s)}{Z_1(s) + Z_2(s)} V_o(s)$$

$$\Rightarrow \frac{V_o(s)}{V_i(s)} = \frac{A}{1 + AZ_1(s)/(Z_1(s) + Z_2(s))}$$

For large A , we disregard unity in the denominator and it becomes:

$$\frac{V_o(s)}{V_i(s)} = \frac{Z_1(s) + Z_2(s)}{Z_1(s)}$$



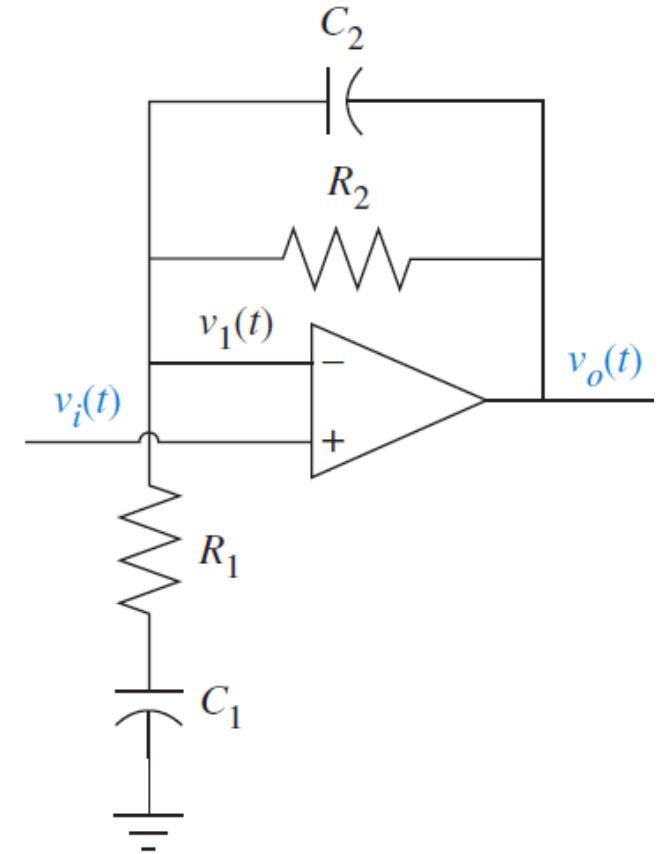
Noninverting Operational Amplifiers

E.g. Find the transfer function.

$$Z_1(s) = R_1 + \frac{1}{C_1 s}$$

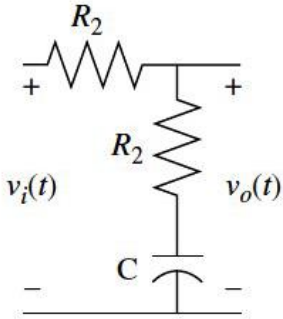
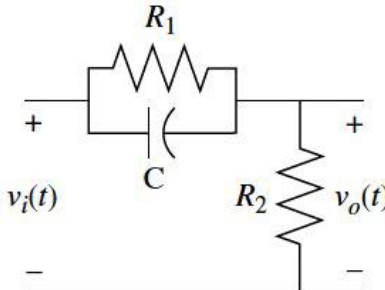
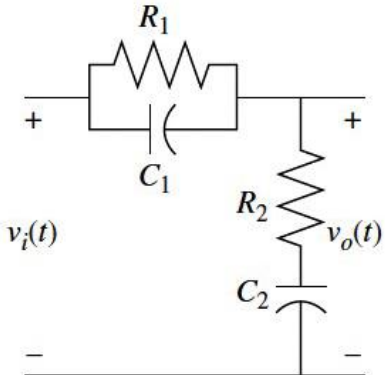
$$Z_2(s) = \frac{R_2(1/C_2 s)}{R_2 + (1/C_2 s)}$$

$$\frac{V_o(s)}{V_i(s)} = \frac{C_2 C_1 R_2 R_1 s^2 + (C_2 R_2 + C_1 R_2 + C_1 R_1)s + 1}{C_2 C_1 R_2 R_1 s^2 + (C_2 R_2 + C_1 R_1)s + 1}$$



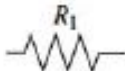
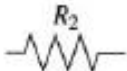
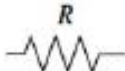
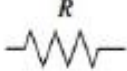
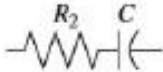
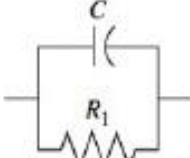
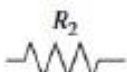
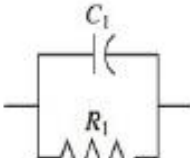
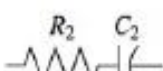
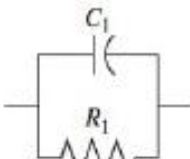
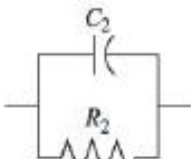
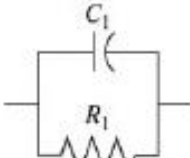
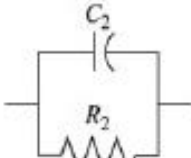
Controller realizations by means of passive circuits

TABLE 9.11 Passive realization of compensators

Function	Network	Transfer function, $\frac{V_o(s)}{V_i(s)}$
Lag compensation		$\frac{R_2}{R_1 + R_2} \frac{s + \frac{1}{R_2 C}}{s + \frac{1}{(R_1 + R_2)C}}$
Lead compensation		$\frac{s + \frac{1}{R_1 C}}{s + \frac{1}{R_1 C} + \frac{1}{R_2 C}}$
Lag-lead compensation		$\frac{\left(s + \frac{1}{R_1 C_1}\right) \left(s + \frac{1}{R_2 C_2}\right)}{s^2 + \left(\frac{1}{R_1 C_1} + \frac{1}{R_2 C_2} + \frac{1}{R_2 C_1}\right) s + \frac{1}{R_1 R_2 C_1 C_2}}$

Controller realizations by means of active circuits – using an inverting OPAMP

TABLE 9.10 Active realization of controllers and compensators, using an operational amplifier

Function	$Z_1(s)$	$Z_2(s)$	$G_c(s) = -\frac{Z_2(s)}{Z_1(s)}$
Gain			$-\frac{R_2}{R_1}$
Integration			$-\frac{1}{RCs}$
Differentiation			$-RCs$
PI controller			$-\frac{R_2}{R_1} \left(s + \frac{1}{R_2 C} \right)$
PD controller			$-R_2 C \left(s + \frac{1}{R_1 C} \right)$
PID controller			$-\left[\left(\frac{R_2}{R_1} + \frac{C_1}{C_2} \right) + R_2 C_1 s + \frac{1}{s} \right]$
Lag compensation			$-\frac{C_1}{C_2} \frac{\left(s + \frac{1}{R_1 C_1} \right)}{\left(s + \frac{1}{R_2 C_2} \right)}$ where $R_2 C_2 > R_1 C_1$
Lead compensation			$-\frac{C_1}{C_2} \frac{\left(s + \frac{1}{R_1 C_1} \right)}{\left(s + \frac{1}{R_2 C_2} \right)}$ where $R_1 C_1 > R_2 C_2$

Q3-1

PROBLEM: Find the transfer function, $G(s) = V_L(s)/V(s)$,

ANSWER: $V_L(s)/V(s) = (s^2 + 2s + 1)/(s^2 + 5s + 2)$

